

SMART FISHER LANKA

Monorepo containing:

- `Backend/` — NestJS + MongoDB API
- `web-app/` — Next.js admin/web frontend
- `mobile/` — Expo React Native mobile app
- `model/` — Python scripts for model training/inference

Prerequisites

Required

- Node.js 18+ (recommended)
- `pnpm` (recommended, because this repo includes `pnpm-lock.yaml` files)
- MongoDB (Atlas or local)

Mobile development (only if running the Expo app)

- Expo Go app (quickest) OR Android Studio / Xcode for emulators

Model scripts (only if running ML scripts)

- Python 3.10+ and `pip`

Install dependencies

This repo has separate apps, so install dependencies per folder.

```
```bash
Backend (NestJS)
pnpm install

Web app (Next.js)
pnpm install

Mobile app (Expo)
pnpm install
```
```

Environment variables

Backend (`Backend/.env`)

Create `Backend/.env`:

```
``env
# MongoDB connection string
MONGO=mongodb+srv://<user>:<pass>@<cluster>/<db>?
retryWrites=true&w=majority

# JWT secret used to sign tokens
JWT_SECRET=change-me

# Optional (defaults to 5000)
PORT=5000

# =====
# JWT AUTHENTICATION (REQUIRED)
# =====
JWT_SECRET="generate-a-secure-random-string-min-32-chars"
JWT_EXPIRES_IN=1h
JWT_REFRESH_SECRET="generate-another-secure-random-string-min-32-
chars"
JWT_REFRESH_EXPIRES_IN=7d
``
```

Notes:

- The backend sets a global prefix: `/api/v1`.
- By default it runs at `http://localhost:5000`.
- CORS is enabled for local dev origins (see `Backend/src/main.ts`).

Web app (`web-app/.env.local`)

Create `web-app/.env.local`:

```
``env
# Base URL of the backend INCLUDING the /api/v1 prefix
NEXT_PUBLIC_API_BASE_URL=http://localhost:5000/api/v1
``
```

If you do not set this, the app falls back to `http://localhost:5000/api/v1`.

Mobile app (mobile/.env)

The mobile app reads the API base URL from `EXPO\_PUBLIC\_API\_KEY` (name is a bit misleading, but that's what the code uses).

Create `mobile/.env`:

```
```env
For iOS simulator / Android emulator you can often use localhost.
For a real device, use your computer's LAN IP (e.g. http://192.168.1.20:5000/
api/v1)
EXPO_PUBLIC_API_KEY=http://localhost:5000/api/v1
```
```

Run the applications

1) Start the backend API

```
cd apps/backend
```

```
```bash
pnpm -C start:dev
```
```

Backend will log the URL (default `http://localhost:5000`).

Quick check:

- `GET http://localhost:5000/api/v1` (may 404 depending on routes, but the server should be reachable)

2) Start the web app

Frontend (Next.js)
cd apps/frontend

Development mode

```
pnpm run dev
```

Open the URL printed by Next.js (usually `http://localhost:3000`).

3) Start the mobile app

```
cd apps/mobile
```

```
```bash
npx expo start
```
```

Then:

- press `i` for iOS simulator, `a` for Android emulator, or
- scan the QR code in Expo Go (make sure `EXPO\_PUBLIC\_API\_KEY` points to a reachable IP from your phone).

Model (Python) — training & prediction

The ML scripts live under `model/finding fish location/train/`.

Install Python dependencies

There is no `requirements.txt` currently; a typical setup is:

```
```bash
python -m pip install -U pip
python -m pip install pandas numpy scikit-learn joblib global-land-mask
```
```

`global-land-mask` is used to filter land points in Sri Lanka (see `land\_mask.py`).

Train the model

```
```bash
python "model/finding fish location/train/train_random_forest.py" \
 --data "model/finding fish location/train/final_dataset_no_bathymetry.csv"
```
```

This writes a model artifact under:

- `model/finding fish location/train/models/rf\_fish\_zone\_model.pkl`

Predict fish presence (0/1)

```
```bash
```

```
python "model/finding fish location/train/predict_fish_zone.py" \
 --lat 7.2 --lon 80.6 --sst 28.0 --chlorophyll 0.3 --u 0.2 --v 0.1
...
```

## ## Useful scripts

### ### Backend

- Dev server: `pnpm -C Backend start:dev`
- Build: `pnpm -C Backend build`

### ### Web

- Dev server: `pnpm -C web-app dev`
- Production build: `pnpm -C web-app build`

### ### Mobile

- Expo dev server: `pnpm -C mobile start`
- Android: `pnpm -C mobile android`
- iOS: `pnpm -C mobile ios`

## ## Troubleshooting

- **\*\*Web/mobile can't reach the backend\*\***: confirm `NEXT\_PUBLIC\_API\_BASE\_URL` / `EXPO\_PUBLIC\_API\_KEY` is correct and includes `/api/v1`.
- **\*\*Running on a real phone\*\***: do not use `localhost`; use your computer's LAN IP (same Wi-Fi).
- **\*\*MongoDB connection errors\*\***: verify `Backend/.env` has a valid `MONGO` connection string and that your IP is allowed in Atlas.

```
=====
```

### # SECURITY

```
=====
```

```
BCRYPT_SALT_ROUNDS=12
```

```
SESSION_SECRET=your-session-secret-key
```

```
=====
```

### # EMAIL (Notification System)

```
=====
```

**# Set EMAIL\_ENABLED=true to enable email notifications**

EMAIL\_ENABLED=false

EMAIL\_HOST=smtp.gmail.com

EMAIL\_PORT=587

EMAIL\_USER=your-email@gmail.com

EMAIL\_PASSWORD=your-app-password

EMAIL\_SECURE=false

EMAIL\_FROM=noreply@Smartfisher.com

EMAIL\_FROM\_NAME=Smart fisher